ADITYA RAJ

Full-Stack Engineer & Open Source Maintainer

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PROFESSIONAL SUMMARY

Self-taught full-stack engineer, IIT Madras B.S. student, and open-source maintainer. Published 4 Python packages on PyPI and maintain the GUN101 cryptographic protocol suite across multiple security-focused repositories. Build software across cryptography, developer tooling, AI systems, computer vision, and full-stack engineering, with experience designing security-sensitive protocols, AST-based code analysis, and production-oriented web systems.

TECHNICAL SKILLS

Languages: Python, C++, C, TypeScript, JavaScript, Kotlin
Frontend: React, Next.js, Vite, Tailwind CSS, Three.js, D3.js
Backend: FastAPI, Node.js, Express, Flask, Django
Databases: PostgreSQL, Supabase, Firebase Firestore, MongoDB, Redis, SQLite, ChromaDB
AI / ML: Google Gemini, NVIDIA NIM, Groq, Hugging Face, RAG, HMA, LiveKit
Security & Crypto: AES-256-GCM, Argon2id, RSA-4096, PBKDF2, TPM 2.0, JWT, bcrypt, signed QR tokens
Systems / DevOps: Linux, Docker, WebSockets, async job queues, AST parsing, Git, Vercel, Render, Firebase
Open Source: Package publishing, code review, release management, semantic versioning, contributor triage

EDUCATION

B.S. in Data Science and Applications — Indian Institute of Technology Madras
Expected 2029

OPEN SOURCE MAINTAINER

GUN101 Protocol Suite — Maintainer | 3 repositories | 4 PyPI packages

- Maintain three open-source cryptographic libraries: GUN101 for password-based AES-256-GCM encryption, GUN101-GKP for passwordless RSA-4096/AES-256-GCM file encryption, and GUN101-TPM for TPM 2.0 hardware-bound encryption.
- Maintain security-focused engineering infrastructure across the suite, including threat models, security policies, test vectors, contributing guidelines, governance documentation, and reproducible dependency configurations.
- Managed external contributions and coordinated community releases, including contributor PR review, security validation, and PyPI release management; the protocol has undergone external review with documented responses and security documentation.

PROJECTS

GUN101 — Password-Based Cryptographic Library **github.com/dialga-cmd/gun101**

- Built an open-source encryption library combining Argon2id password derivation with AES-256-GCM authenticated encryption for protected data and file workflows.
- Maintained protocol security documentation, threat modeling, test vectors, contribution standards, and release infrastructure as part of the broader GUN101 suite.
- Stack: Python, AES-256-GCM, Argon2id

GUN101-GKP — Ghost Key Protocol **github.com/dialga-cmd/gun101-gkp**

- Built a passwordless asymmetric encryption protocol using RSA-4096 identity keys with AES-256-GCM for encrypted payloads, separating identity authentication from data encryption.
- Maintained the protocol as an open-source PyPI package with documented security considerations, tests, and protocol behavior.
- Stack: Python, RSA-4096, AES-256-GCM

GUN101-TPM — Hardware-Bound Encryption **github.com/dialga-cmd/gun101-tpm**

- Built a TPM 2.0-backed encryption library that binds cryptographic protection to hardware-backed TPM keys, adding a hardware trust layer beyond software-only encryption.
- Implemented and documented the Linux TPM workflow as part of the GUN101 cryptographic protocol suite.
- Stack: Python, TPM 2.0, cryptography

LucidGrasp — C++ Image Search & Computer Vision **github.com/dialga-cmd/LucidGrasp**

- Built a cross-platform image search engine in C++ using OpenCV and Qt6, indexing local image libraries and retrieving visually similar images including edited, recolored, cropped, and rotated variants.
- Combined SSIM, ORB keypoint matching, and HSV color-histogram comparison into a weighted similarity pipeline, with two-stage candidate ranking and background-threaded indexing/search to keep the GUI responsive.
- Stack: C++, OpenCV, Qt6, CMake

CodeLens — AI Codebase Intelligence & 3D Visualization **github.com/dialga-cmd/CodeLens**

- Built a codebase-analysis platform that clones GitHub repositories, parses source with Tree-sitter, extracts file/dependency relationships, detects architectural hotspots, and visualizes repository structure through interactive 3D graphs.
- Integrated an AI analysis layer for repository understanding and contextual codebase queries, with separate Next.js/React frontend and FastAPI/Python backend services.
- Stack: Next.js, React, FastAPI, Python, Tree-sitter, Three.js, React Force Graph, Firebase, Gemini

HMA — Retrieval-Augmented Hierarchy **github.com/dialga-cmd/hma**

- Designed and published a structural alternative to chunk-based RAG that builds a knowledge map of files, imports, configurations, and documentation before routing queries to relevant files and directly connected context.
- Implemented offline project indexing, relationship extraction, LLM-based file routing, single-hop graph traversal, and a Python API/CLI without requiring vector databases or embedding models.

- Stack: Python, AST parsing, LLM APIs, PyMuPDF, python-docx

ACHIEVEMENTS

- Google PromptWar Hackathon — Ranked 310 / 46,000+ with FanWayfinder, an AI-powered FIFA World Cup 2026 stadium navigation assistant.
- OpenAI x Outskill Hackathon 2026 — Selected participant; independently built and presented CodeLens in 2 days.
- Google Gen AI Hackathon — Built Disaster Intelligence Copilot, an AI public-safety decision platform integrating live disaster data, AI report triage, and RAG-grounded safety Q&A.;
- BRICS Track — Built CivicPulse for AI for Digital Public Infrastructure and Governance.
- Published 4 open-source Python packages on PyPI: gun101, gun101-gkp, gun101-tpm, hma.
- GUN101 protocol suite — Managed external contributor activity, reviewed and merged community PRs, and shipped release v2.1.2.